Geometric Limits of Defense Design: Impossibility Theorems for Continuous Prompt Injection Defenses

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Abstract

Prior work on LLM safety has focused on discovering adversarial inputs or bounding perturbation robustness around individual examples. We ask a different question: what are the *geometric limits of defense design itself*?

We prove three results of increasing strength. First, any continuous, utility-preserving defense on a connected Hausdorff space must fix boundary points of the alignment deviation surface (*boundary fixation*). Second, when the defense and alignment function are both Lipschitz, the defense cannot uniformly reduce alignment deviation far below the threshold on a positive-measure band around the boundary (*ε-robust constraint*). Third, under a transversality condition—the alignment surface rises faster than the defense can pull it down—the defense leaves a *positive-measure region genuinely unsafe*: $f(D(x)) > \tau$ persists over volume (*persistent unsafe region*).

These are not statements about attacks or training—they are constraints on what any admissible defense mapping can do, independent of the model, data, or optimization procedure. The Tietze extension theorem ensures that any finite set of behavioral observations admits continuous extensions for which all three impossibilities hold.

We also prove a parallel set of discrete impossibility results that require no topology, and extend the theory to multi-turn interactions (the impossibility recurs at every turn), stochastic defenses (randomization does not help), and capacity parity (utility preservation taxes the defense even under equal resources).

The paper defines the theoretical speed limit for AI safety. It proves we can't build a 'Perfect Shield' in a continuous space. However, by using multi-layered, discontinuous defenses and ensuring our boundary behavior is benign, we can operate safely within those limits. The 'Manifold of Failure' exists, but we can ensure it's empty of actual harm. The full theory is machine-verified in Lean 4 with Mathlib (37 files, ∼300 theorems, zero `sorry`, three standard axioms) and validated against empirical findings from three frontier LLMs. `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities?

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Most current work on LLM safety implicitly assumes the answer is yes—at least in principle, even if current tools fall short in practice. The defense mechanisms people have proposed span a wide range: input classifiers that flag suspicious prompts [1], output filters that catch unsafe completions [9], and constitutional rewriting pipelines that reshape model behavior [3]. But despite their differences, all of these defenses share the same underlying structure: they are functions $D\colon X \to X$ that preprocess prompts before the model sees them, aiming to map unsafe inputs to safe equivalents while leaving safe inputs alone.

Prior work has largely focused on discovering or mitigating individual adversarial inputs. We ask a different question: what are the *geometric limits of the defense mapping itself*? We show that under mild and commonly assumed design constraints—continuity and utility preservation—no defense can eliminate the safety boundary. Vulnerabilities arise not from specific inputs, but from the structure of the space and the constraints imposed on admissible defenses. This shifts the problem from adversarial search to geometric inevitability.

We prove three results of increasing strength.

Tier 1: Boundary fixation. If the defense is continuous, utility-preserving, and the prompt space is connected, then the defense must fix at least one boundary point—a prompt where alignment deviation equals the threshold exactly—passing it through with no remediation (Theorem 4.1). The proof is five steps of point-set topology.

Tier 2: ε-robust constraint. If the defense is also K-Lipschitz and the alignment deviation function is L-Lipschitz, the defense cannot uniformly reduce alignment deviation far below τ on a neighborhood of the fixed boundary point. For any x within distance δ of the fixed point z:

$$f(D(x)) \;\geq\; \tau - L(K+1)\,\delta. \tag{1}$$

The ε-band $\{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure for every $\varepsilon > 0$ (Theorem 5.2). This constrains the *depth* to which the defense can push near-boundary points into the safe region, though it does not by itself prevent the defense from nudging them slightly below τ.

Tier 3: Persistent unsafe region. Under a transversality condition—the alignment surface has directional slope exceeding $L(K+1)$ at the boundary—the defense leaves a *positive-measure region genuinely unsafe*:

$$f(D(x)) > \tau \quad \text{for all } x \in \mathcal{S}, \qquad \mu(\mathcal{S}) > 0. \tag{2}$$

This is the strongest result: not boundary fixation, not a near-threshold band, but genuine persistent unsafe volume (Theorem 6.2). The condition holds whenever the alignment surface rises faster than the defense can pull it down—empirically observed in all three models we study.

Why continuity? Continuity captures a natural requirement for practical defenses: small changes in input should not produce arbitrarily large changes in the defense's output, as such instability would degrade usability and reliability. Any defense based on learned representations, smooth transformations, or differentiable processing satisfies this constraint. Discontinuous defenses (hard blocklists, keyword filters) escape the theorem but suffer their own well-known failure modes.

From discrete to continuous. All three results apply to continuous interpolants of discrete data. The Tietze extension theorem guarantees that any finite set of behavioral observations in a normal space admits continuous extensions, and the impossibilities hold for every such extension

(Theorem 7.1). This does not claim the results hold for the underlying discrete token system directly, but for every continuous model consistent with the observations.

Both the basic and ε-robust proofs are machine-verified in Lean 4 with zero `sorry` statements, as is the persistent unsafe region theorem. All three conditions—continuity, utility preservation, and connectedness—are individually necessary; we give counterexamples for each.

Contributions.

- **Tier 1**: Boundary fixation for continuous utility-preserving defenses (Theorem 4.1).
- **Tier 2**: ε-robust constraint showing the defense cannot uniformly reduce AD far below τ on a positive-measure band (Theorems 5.1 and 5.2).
- **Tier 3**: Persistent unsafe region under transversality (Theorem 6.2).
- A formally verified bridge from discrete data to continuous theory via Tietze extension (Theorem 7.1).
- Direct discrete impossibility results requiring no topology (Section 8).
- Multi-turn, stochastic, and capacity parity extensions (Section 9).
- Counterexamples proving each hypothesis is necessary (Section 4.1).
- Supporting theorems: basin structure, fragmentation, convergence, transferability, authority monotonicity, gradient ascent, fine-tuning stability, cost asymmetry (Sections 10 to 13).
- Theory–experiment correspondence with three LLMs (Section 14).
- Lean 4 artifact: 37 files, $\sim$300 theorems, zero `sorry`, three standard axioms (Section 16).

2 Related Work

Adversarial robustness. The existence of adversarial examples in neural networks has been known for over a decade [15, 8], and a substantial body of work has pursued formal verification of robustness properties—through abstract interpretation [14], SMT-based methods [10], and randomized smoothing [7]. Unlike adversarial robustness results, which characterize the existence of perturbations or bound robustness radii around individual inputs under specific norms, our result is a *structural constraint on the defense mapping itself*, independent of perturbation models, norms, or training procedures. Certified defenses provide guarantees about perturbation neighborhoods; our result constrains the *global behavior* of the defense function, showing that no such defense can eliminate boundary regions.

Prompt injection and jailbreaking. Techniques for finding individual vulnerabilities include gradient-based attacks like GCG [18], search-based approaches like PAIR [6] and TAP [11], and quality-diversity methods like Rainbow Teaming [13]. The Manifold of Failure framework [4] uses MAP-Elites to map the *continuous topology* of failures. Existing work identifies individual failure cases or distributions of attacks. Our result shows that these failures are not incidental but *topologically unavoidable* under mild assumptions, providing a formal explanation for their persistence.

Impossibility results in ML. The no-free-lunch theorems [17] limit average performance across tasks. The robustness–accuracy tradeoff literature [16] shows that adversarial robustness costs standard performance. Our result is different in character: it holds *pointwise* for any continuous defense on a *fixed task*, with no dependence on distributions, learning algorithms, or training data.

Topological views of neural networks. Work on decision boundary geometry, the manifold hypothesis, and topology of neural representations has established that topological structure mat-

ters. While our proof relies on elementary topology, the novelty lies in identifying that *standard design constraints for LLM defenses induce a forced fixed-point structure at the safety boundary*—a connection not previously characterized. The persistent unsafe region theorem (Tier 3) goes further, using metric structure to establish volumetric persistence.

Formal verification of ML systems. Proof assistants like Lean and Coq have been applied to verify properties of neural networks [2] and optimization algorithms [5]. To the best of our knowledge, ours is the first formally verified impossibility result specifically targeting prompt injection defense.

3 Setup

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space. An* alignment deviation function *is a continuous map $f\colon X \to \mathbb{R}$. Given a threshold $\tau \in \mathbb{R}$:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{3}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{4}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{5}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$, and* complete *if $f(D(x)) < \tau$ for all $x \in X$.*

4 Tier 1: Boundary Fixation

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space. Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Let $D\colon X \to X$ be continuous with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$. This set is nonempty.

Proof. Five steps, each using exactly one hypothesis.

Step 1 (Fixed-point set is closed). $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under $x \mapsto (D(x), x)$. Hausdorff $\Rightarrow \Delta$ closed $\Rightarrow \mathrm{Fix}(D)$ closed.

Step 2 (Closure of safe region $\subseteq$ fixed points). $S_\tau \subseteq \mathrm{Fix}(D)$ by utility preservation. $\mathrm{Fix}(D)$ closed $\Rightarrow \overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If also closed, it would be clopen, hence $\emptyset$ or X—contradicting both regions nonempty.

Step 4 (Boundary point in closure). S_τ not closed $\Rightarrow \overline{S_\tau} \supsetneq S_\tau$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) \leq \tau$ (closure $\subseteq \{f \leq \tau\}$) and $f(z) \geq \tau$ ($z \notin S_\tau$). So $f(z) = \tau$.

Step 5 (Defense fixes it). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau \not< \tau$. □

Corollary 4.2. *No continuous utility-preserving defense is complete.*

4.1 Each Hypothesis Is Necessary

Counterexample 4.3 (Removing connectedness)**.** *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample 4.4 (Removing continuity)**.** *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample 4.5 (Removing utility preservation). *$D(x) = x_0$ for a fixed safe point: continuous and complete, but destroys all inputs.*

Remark 4.6 (The defense trilemma). *A defense can satisfy at most two of: continuity, utility preservation, completeness.*

Remark 4.7 (Not all boundary points are fixed). *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points isolated from the safe side may escape. Verified in Lean with $f(x) = x^2$, $\tau = 0$.*

5 Tier 2: ε-Robust Constraint

Tier 1 fixes boundary points—potentially measure zero. Tier 2 extends the constraint to a positive-measure neighborhood.

Theorem 5.1 (ε-Robust Defense Constraint). *Let X be a connected Hausdorff metric space, f be L-Lipschitz with $S_\tau, U_\tau \neq \emptyset$, and D be K-Lipschitz with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$, $D(z) = z$, such that for all $x \in X$ (with strongest constraint near z):

$$f(D(x)) \;\geq\; \tau \;-\; L(K+1)\,\mathrm{dist}(x, z). \tag{6}$$

Proof. From Theorem 4.1, z exists. Since D is K-Lipschitz and $D(z) = z$: $\mathrm{dist}(D(x), x) \leq (K+1)\,\mathrm{dist}(x, z)$ (triangle inequality). Since f is L-Lipschitz: $|f(D(x)) - f(x)| \leq L(K+1)\,\mathrm{dist}(x, z)$. Extracting the lower bound gives $f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x, z) \geq \tau - L(K+1)\,\mathrm{dist}(x, z)$ (using $f(z) = \tau$ and Lipschitz of f). □

Theorem 5.2 (Positive-Measure ε-Band). *Under the hypotheses of Theorem 5.1, for every $\varepsilon > 0$ such that f takes values below $\tau - \varepsilon$, the ε-band $\mathcal{B}_\varepsilon = \{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure (under any measure positive on nonempty open sets). Specifically,* $\mathrm{ball}(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$ *for any c with $f(c) = \tau - \varepsilon/2$.*

Remark 5.3 (What Tier 2 does and does not say). *Theorem 5.1 constrains the* depth *to which the defense can push near-boundary points below τ. It does* not *prove these points remain unsafe—the defense may map them to values in $[\tau - L(K+1)\delta,\ \tau)$, which is below threshold. The bound is tight: a defense mapping every point in a δ-ball to f-value $\tau - L(K+1)\delta$ would saturate it.*

6 Tier 3: Persistent Unsafe Region

Tier 2 constrains depth but permits the defense to nudge points below τ. Under a transversality condition, Tier 3 proves the defense *cannot make certain points safe at all.*

Definition 6.1 (Steep region). *Given a fixed boundary point z (from Theorem 4.1), define*

$$\mathcal{S} = \{x \in X : f(x) > \tau + L(K+1)\ \mathrm{dist}(x, z)\}.$$

This is the set of points where the alignment deviation exceeds τ by more than the defense's Lipschitz bound can compensate.

Theorem 6.2 (Persistent Unsafe Region). *Under the hypotheses of Theorem 5.1, if $\mathcal{S}$ is nonempty, then:*

1. *$\mathcal{S}$ is open.*
2. *$\mathcal{S}$ has positive measure (under any measure positive on nonempty open sets).*

3. For every $x \in \mathcal{S}$: $f(D(x)) > \tau$.
In particular, $\mu(\{x : f(D(x)) > \tau\}) > 0$—the defense leaves a positive-measure region genuinely unsafe.

Proof. **(1)** $\mathcal{S}$ is the strict superlevel set of the continuous function $x \mapsto f(x) - L(K+1)\,\mathrm{dist}(x, z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + L(K+1)\,\mathrm{dist}(x, z)$. From the proof of Theorem 5.1, the defense satisfies $f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x, z)$. Combining:

$$f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x, z) > \tau.$$

The defense cannot bring these points below threshold. □

Remark 6.3 (When is $\mathcal{S}$ nonempty?)**.** *The set $\mathcal{S}$ is nonempty whenever the alignment surface rises with directional slope exceeding $L(K+1)$ at the boundary point z. In a normed space, if f has directional derivative $c > L(K+1)$ along some unit vector v at z, then $z + tv \in \mathcal{S}$ for small $t > 0$ (verified in Lean as* `transversality_from_deriv`*).*

Informally: if the alignment surface rises faster than the defense can pull it down, unsafe behavior persists over volume. This condition is empirically observed in all three models we study—the Lipschitz constant of the alignment surface near boundary crossings consistently exceeds reasonable defense aggressiveness bounds.

Remark 6.4 (The defense's fundamental dilemma)**.** *The three tiers reveal a progressive bind. Tier 1: the defense* must *fix boundary points. Tier 2: it cannot push nearby points* deep *into safety. Tier 3: if the landscape is steep, it cannot push them into safety* at all*. Increasing K (defense aggressiveness) raises the transversality threshold $L(K+1)$, requiring an even steeper landscape to trigger Tier 3—but it also makes the defense less stable and widens the Tier 2 constraint band. The defense cannot escape the geometry.*

7 From Discrete Data to Continuous Theory

Theorem 7.1 (Continuous Relaxation)**.** *Let X be a connected, normal, Hausdorff space. Let $S \subset X$ be a finite set of behavioral observations with $g(p) < \tau$ for some p and $g(q) > \tau$ for some q.*

Then there exists a continuous function $f\colon X \to \mathbb{R}$ agreeing with all observations for which Theorems 4.1 and 5.1 hold. If f is additionally Lipschitz (as it is for GP posterior means under standard kernel regularity assumptions, e.g., jointly continuous kernel on a compact domain), Theorem 6.2 applies wherever the transversality condition is met.

Proof. Finite subsets of T_1 spaces are closed. By the Tietze extension theorem, g extends to a continuous f agreeing at every sample point. Since $f(p) < \tau < f(q)$, Theorem 4.1 applies. Lipschitz extensions (which exist by McShane–Whitney) enable Theorems 5.1 and 6.2. □

Remark 7.2. *The interpolant is not unique. All three impossibilities hold for* every *continuous extension consistent with the data. This does not claim the results hold for the underlying discrete token system directly, but for every continuous model consistent with the observations.*

Corollary 7.3. *Theorem 7.1 applies in $\mathbb{R}^n$ for every $n \geq 1$.*

8 Direct Discrete Impossibility

The Tietze bridge connects discrete data to continuous theory, but a natural objection remains: perhaps the impossibility is an artifact of the continuous relaxation, not a property of the discrete system. To address this, we prove a parallel set of results that work directly on finite sets with no topology, no continuity, and no Tietze extension. These are verified in Lean as `MoF_12_Discrete`.

Theorem 8.1 (Discrete IVT). *Let $f\colon \{0,\ldots,n{+}1\} \to \mathbb{R}$ with $f(0) < \tau$ and $f(n{+}1) \geq \tau$. Then there exists i with $f(i) < \tau$ and $f(i{+}1) \geq \tau$.*

Proof. By induction: if no consecutive crossing exists and $f(0) < \tau$, then $f(k) < \tau$ for all k, contradicting $f(n{+}1) \geq \tau$. □

This is the discrete analogue of boundary existence. The next result is the discrete analogue of boundary fixation.

Theorem 8.2 (Discrete Defense Fixation). *If D fixes all safe vertices ($f(v) < \tau \Rightarrow D(v) = v$), the boundary crossing from Theorem 8.1 has its safe endpoint fixed: $D(i) = i$.*

The proof is immediate: $f(i) < \tau$, so $D(i) = i$ by utility preservation. No continuity is needed—just the identity constraint on safe inputs.

Where the continuous and discrete paths diverge is in what replaces continuity as the bottleneck. In the continuous setting, the defense is limited by its Lipschitz constant. In the discrete setting, it is limited by its *capacity*—the number of distinct output behaviors it can realize.

Theorem 8.3 (Capacity Exhaustion). *If there are m adversarial configurations and the defense can realize fewer than m distinct behaviors, at least two configurations receive the same treatment—so at least one is mishandled.*

Proof. Pigeonhole principle, verified in Lean via `Fintype.exists_ne_map_eq_of_card_lt`. □

Theorem 8.4 (Exponential Attack Surface). *The attack surface grows as $2^d \geq d{+}1$. For any fixed budget B, there exists d_0 such that $2^d > B$ for all $d \geq d_0$. No fixed-capacity defense suffices against an exponentially growing configuration space.*

Together, these results form a *discrete defense trilemma*: on any finite ordered set where f crosses τ, a utility-preserving defense must leave the safe side of the crossing fixed, and a bounded-capacity defense must eventually be overwhelmed by the attack surface. The continuous and discrete paths arrive at the same conclusion by different routes.

9 Extensions: Multi-Turn, Stochastic, and Capacity Parity

A natural objection to the preceding results is that they assume a static, deterministic, single-turn defense. Real systems are multi-turn, stochastic, and may have capacity comparable to the attacker. We show that the impossibility survives all three extensions.

9.1 Multi-Turn Impossibility

Modern LLMs maintain state across conversation turns. One might hope that a defense could use accumulated context to progressively eliminate boundary vulnerabilities. This does not work.

Theorem 9.1 (Multi-Turn Impossibility). *Let $\{f_t, D_t\}_{t=1}^{T}$ be a sequence of alignment functions and defenses over T turns, where each f_t is continuous with safe and unsafe regions, and each D_t is continuous and utility-preserving. Then for* every *turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Proof. Apply Theorem 4.1 to (f_t, D_t) at each turn. The functions f_t and D_t may depend on the full history of previous turns—this does not matter. At each timestep, the defense faces a fresh instance of boundary fixation on the current alignment surface. □

The impossibility does not accumulate or diminish over turns—it simply recurs. Memory, context, and adaptive strategies do not change the topology of the current turn's alignment surface. But the multi-turn setting is worse than "the same problem repeated." The attacker gains several structural advantages.

Information accumulation. At each turn, the defense's response reveals whether the probed input is safe, unsafe, or on the boundary. Over T turns with distinct probes, the attacker classifies T points, progressively narrowing the boundary's location. The defense leaks information; the attacker consumes it.

Monotone attacker improvement. The attacker's best observed alignment deviation score across turns is non-decreasing: the running maximum of a sequence never drops (verified in Lean as `running_max_monotone`). Each turn either improves the attacker's best known exploit or leaves it unchanged.

Steering toward transversality. If the attacker can influence the alignment surface—for instance, by crafting conversation context that shifts the landscape—and the directional slope at the boundary varies continuously with the attacker's influence parameter α, then the intermediate value theorem guarantees the existence of α^* at which the slope crosses the transversality threshold $L(K+1)$. For $\alpha > \alpha^*$, Tier 3 applies: persistent unsafe volume. The attacker can find α^* by binary search (verified in Lean as `transversality_reachable`).

No progressive safety. As long as both safe and unsafe regions exist at each turn, every turn produces a boundary fixed point—no turn is "free." The only escape is to make $U_\tau = \emptyset$ at some turn, which requires changing the alignment surface f_t itself (e.g., via in-context learning that suppresses all unsafe behavior), not merely preprocessing inputs through D_t.

9.2 Stochastic Defense Impossibility

If the defense is randomized—mapping each input to a distribution over outputs rather than a fixed point—one might hope that the boundary fixation dissolves in expectation.

Theorem 9.2 (Stochastic Defense Impossibility). *Let $f\colon X \to \mathbb{R}$ be continuous on a connected Hausdorff space with $S_\tau, U_\tau \neq \emptyset$. Let D be a stochastic defense, and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$ (the expected alignment deviation after defense). If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$ (utility preservation in expectation), then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Proof. The function $h = g - f$ is continuous and vanishes on S_τ. Since S_τ is dense in $\overline{S_\tau}$ and $\{h = 0\}$ is closed, h vanishes on $\overline{S_\tau}$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ (by the argument of Theorem 4.1) and $g(z) = f(z) = \tau$. The defense cannot reduce even the *expected* alignment deviation below τ at these points. □

Randomization does not help: the expected alignment deviation after a stochastic defense inherits the same boundary fixation as a deterministic one.

9.3 Capacity Parity

Even when the defense has the same computational capacity as the attacker, utility preservation acts as a structural tax.

Theorem 9.3 (Capacity Parity Disadvantage). *Let the defense and attacker both operate on a space of N inputs, of which $|S_\tau|$ are safe. The defense must implement identity on all safe inputs, consuming $|S_\tau|$ of its capacity. The attacker faces no such constraint. Under capacity parity ($C = N$), the defense's effective free capacity is $N - |S_\tau| < N$, giving the attacker a permanent edge of $|S_\tau|$ capacity units.*

This is a counting argument: utility preservation is not free. Every safe input the defense must leave untouched is one fewer degree of freedom available for remapping unsafe inputs. The attacker, who has no obligation to preserve anything, retains full capacity.

10 The Vulnerability Landscape

Theorem 10.1 (Basin Structure). *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure that assigns positive mass to nonempty open sets (e.g., Lebesgue measure on $\mathbb{R}^n$), U_τ has positive measure.*

Theorem 10.2 (Basin Fragment Minimum Size). *If f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

11 Attack Properties

Theorem 11.1 (Perturbation Robustness). *If f is L-Lipschitz and $f(p) > \tau$, then* $\mathrm{ball}(p, (f(p) - \tau)/L) \subseteq U_\tau$. *The radius is monotone in $f(p)$.*

Theorem 11.2 (Iterative Convergence). *Any monotone-improvement operator T with score in* $[0, 1]$ *converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem 11.3 (Transferability). $\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. *Transfer costs zero additional queries.*

Theorem 11.4 (Authority Monotonicity). *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, then the vulnerability set $\{x : f(y, x) > \tau\}$ is upward-closed and a critical threshold $a_2^*(y)$ exists (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem 11.5 (Gradient Ascent). *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

Table 1: Falsifiable predictions confirmed by empirical data.

Theorem	Predicts	Confirmed by
Basin Structure (10.1)	Basins are open with positive measure	Heatmaps show extended regions
Fragmentation (10.2)	Smooth → large basins; rough → mosaic	Llama: mesa; GPT-OSS: mosaic
Convergence (11.2)	Attacks exhibit monotone convergence	Convergence curves plateau
Transferability (11.3)	Similar surfaces → shared basins	Llama .93 → GPT-OSS .73 → Mini .47
Authority (11.4)	Horizontal banding	Bands at $a_2 \approx$ .25–.35, .65–.85
Persistent (6.2)	Steep boundaries → unsafe volume persists	Llama's .93 plateau immune to defense
Interior Stability (12.1)	Deep basin points survive fine-tuning	Vulnerability persists across variants
Cost (13.1)	2D tractable, high-d intractable	15K queries fill 63% at d=2

12 Stability Under Fine-Tuning

Theorem 12.1 (Interior Stability). *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem 12.2 (Crossing Preservation). *If f crosses τ transversally on $[a, b]$ and g is continuous with $\|f - g\|_\infty \leq \varepsilon$, then g also crosses τ on $[a, b]$.*

Theorem 12.3 (Patching Is Nonlocal). *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points. (Verified by explicit construction in Lean.)*

13 Cost Asymmetry

Theorem 13.1 (Exponential Cost Asymmetry for Grid-Based Defense). *For a defense monitoring the behavioral space via a grid with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); grid-based defense cost $= N^d$ (exponential in d); the ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$. Verified by `norm_num` in Lean.

14 Experimental Validation

We validate against the Manifold of Failure framework [4], which maps three LLMs over a 2D behavioral space (query indirection × authority framing).

Llama-3-8B (mean AD 0.93, basin rate 93.9%): near-flat mesa, small L, large robustness radii. The directional slope of the alignment surface at boundary crossings far exceeds $L(K+1)$ for any reasonable K, placing this model squarely in Tier 3 territory: the defense provably cannot eliminate the vulnerability volume.

GPT-OSS-20B (mean AD 0.73, basin rate 64.3%): rugged landscape, large L, small fragments. Horizontal bands confirm authority monotonicity. The fragmented boundary presents many transversal crossings, each generating a persistent unsafe pocket.

GPT-5-Mini (peak AD 0.50, basin rate 0%): at $\tau = 0.50$, $U_\tau = \emptyset$, so none of the three tiers apply—correctly predicting no impossibility. The model's strategy: make the boundary harmless rather than eliminate it.

Quantitative checks. Robustness radius: $L \approx 5$ for Llama gives $r = (0.93-0.50)/5 = 0.086$, covering $\sim$2 grid cells—consistent with heatmap smoothness. Cost ratio: $0.01 \times 625 = 6.25$ theoretical vs. $\sim$4 empirical.

15 The Engineering Prescription

The impossibility results do not say that defense is futile—they say that *perfect* defense is impossible under the stated constraints. The gap between impossibility and practice is where engineering lives. The following prescriptions are ordered from most to least actionable.

1. Make the boundary shallow. Set τ so that boundary-level behavior is benign. If $f(z) = \tau$ corresponds to a polite refusal rather than harmful compliance, the impossibility is mathematically true but practically harmless. GPT-5-Mini exemplifies this: its ceiling at AD = 0.50 means the "Manifold of Failure" exists but contains no actual harm.

2. Reduce the Lipschitz constant. Smaller L makes the alignment surface smoother and more predictable. It also raises the transversality threshold $L(K+1)$ for Tier 3: a smoother surface is harder for the attacker to steer into persistent unsafe territory. The tradeoff is that smoother surfaces spread vulnerabilities over wider regions, but those regions are more easily monitored.

3. Reduce the effective dimension. Defense cost grows as N^d. Constraining the prompt interface—standardized formats, restricted API parameters, bounded context lengths—reduces d, making the behavioral space tractable.

4. Monitor, don't eliminate, the boundary. Theorem 12.2 shows transversal crossings persist under fine-tuning, and Theorem 9.1 shows they recur every turn. Rather than attempting the impossible (eliminating boundary points), deploy runtime monitoring that detects approach to the boundary. The Lipschitz bound (Theorem 11.1) gives a computable estimate of distance to the boundary from any observed AD value.

5. Abandon stateful security illusions. Theorem 9.1 proves that accumulating context across turns does not eliminate boundary fixation—the impossibility recurs fresh at every turn. Defenses that rely on "learning" from conversation history to progressively close vulnerabilities are fighting topology, not winning. Each turn is a new instance of the same problem.

6. Do not pay the stochastic tax. Theorem 9.2 proves that randomized defenses—stochastic routing, temperature-based sampling, random prompt perturbation—fail in expectation. The expected alignment deviation after a stochastic defense inherits the same boundary fixation as a deterministic one. Compute spent on randomization is compute wasted, at least for the purpose of eliminating boundary points.

7. Break capacity parity with architectural separation. Theorem 9.3 proves that under equal resources, the defense loses capacity to utility preservation while the attacker retains full freedom. The implication: a defense that shares the same architecture and parameter budget as the model it protects starts at a structural disadvantage. To break this asymmetry, separate the defense from the utility pipeline entirely—for example, by using a dedicated lightweight classifier that does not need to preserve utility (it only flags), paired with a separate response generator that handles safe inputs. This sacrifices the "single continuous wrapper" model but sidesteps the capacity tax.

16 The Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0. The artifact comprises 37 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust + persistent unsafe region (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic + capacity parity (MoF_13_MultiTurn)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)

Every theorem depends on exactly three axioms: `propext`, `Classical.choice`, `Quot.sound`—Lean's standard foundations. No additional axioms.

Key Mathlib lemmas used:

Lean name	**Role**
`isClosed_diagonal`	Hausdorff ⇒ diagonal closed
`LipschitzWith.dist_le_mul`	Lipschitz distance bound
`isClopen_iff`	Connected ⇒ no proper clopen
`closure_minimal`	Closure in closed superset
`IsOpen.measure_pos`	Open nonempty ⇒ positive measure
`intermediate_value₂`	IVT on connected spaces

17 Limitations

Tier 1 is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless.

Tier 2 constrains depth, not direction. The defense may push near-boundary points slightly below τ. The bound limits how far, not whether.

Tier 3 requires transversality. The persistent unsafe region exists only where the alignment surface is steep ($c > L(K + 1)$). Flat or gently sloping boundaries escape Tier 3 (though not Tiers 1–2).

The Tietze bridge applies to interpolants. The impossibilities hold for every continuous extension of discrete data, not for the underlying discrete system itself.

Grid-based cost asymmetry. Theorem 13.1 assumes exhaustive grid enumeration. Learning-based defenses that generalize across the space may sidestep the exponential bound.

Single-turn only. The manifold may shift across conversation turns.

18 Conclusion

We have established three impossibility results for prompt injection defense, forming a progressive hierarchy. Tier 1 (boundary fixation): any continuous, utility-preserving defense must fix boundary points. Tier 2 (ε-robust constraint): the defense cannot uniformly reduce alignment deviation far below τ on a positive-measure band. Tier 3 (persistent unsafe region): under transversality, the defense leaves positive-measure volume genuinely unsafe—$f(D(x)) > \tau$ persists.

The key insight is not about attacks or training but about the *geometry of the defense itself*: continuity and utility preservation together create an anchor at the boundary that constrains what the defense can do in its neighborhood, and metric structure (Lipschitz bounds) makes this constraint quantitative and volumetric.

The discrete-continuous gap is addressed via Tietze extension and a parallel set of purely discrete results (Section 8). The impossibility extends to multi-turn interactions (recurs every turn), stochastic defenses (randomization does not help), and capacity parity (utility preservation taxes the defense even with equal resources). Supporting theorems show that vulnerabilities come in basins, attacks converge and transfer, vulnerabilities survive fine-tuning, and defense cost grows exponentially. Each result is validated against empirical data from three frontier LLMs.

The practical prescription: make the boundary shallow (raise τ), smooth (reduce L), and low-dimensional (constrain the interface). GPT-5-Mini's ceiling at AD = 0.50 shows this working in practice.

The full theory is machine-verified in Lean 4: 37 files, approximately 300 theorems, zero `sorry`, three standard axioms.

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